

Nikhil Kalidasu

University of Texas at Austin (GPA: 3.6)

BS, Computer Science; BS, Biology

Portfolio: [nik875.github.io](https://github.com/nik875)

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Coursework: Data Structures, Algorithms & Complexity, Software Engineering, Operating Systems, Genetics, Data Science, Cell Biology, Randomized Algorithms, Computer Vision, Statistics & Probability, Cloud Computing.

Skills: Python, R, C/C++, Java, JavaScript, PyTorch, TensorFlow, CNN, GNN, Transformers, LLMs, Linux, Bash, NumPy, SQL, Postgres, Pandas, REST APIs, Docker, AWS, GCP, Reinforcement Learning (RL), Bayesian Statistics, HPC.

Work Experience

Collegiate Cyber Defense Competition – Infrastructure & Security

(Jan 2026 – Apr 2026)

- Managed **15+ machines**, maintained SQL/Postgres databases, REST APIs, Docker containers, AWS infrastructure.
- Won **first place** for service uptime and red team suppression and **second place** overall.

Sessions Lab, Cal Tech – Undergraduate Research Assistant

(Jun 2025 – Jan 2026)

- NRIQ: **Bayesian optimization** for predicting kinetic isotope effects in microbial sulfate reduction, TCA cycle.
- Implemented **parallelization** for 28-core HPC, evaluated NRIQ against baselines, developed **user-friendly UI**.

AeroParagon (Counter-Drone Startup) – AI/ML Engineering Intern (Computer Vision)

(Jan 2025 – Mar 2025)

- Fine-tuned RetinaNet CNN**, optimizing with **Neural Architecture Search (NAS)** for edge deployment.
- Productionized auto-labeling pipeline** for 3 GB videos via motion detection with **MoG** and **Kalman filter**.

De Anda Lab, University of Florida – Computational Scientist

(Aug 2024 – Aug 2025)

- Developed [SeREGen](#), machine learning embedding generator for DNA built on **Transformer, CNN, PyTorch**.
 - Used to visualize **40 novel bacterial phyla** and **COVID-19 spike protein** mutations for vaccine resistance.

ECLAIR Robotics – Project Lead

(Aug 2023 – Apr 2025)

- Led a ~25 person team** to develop an autonomous RC car from scratch, self-driving algorithm runs on **Raspberry Pi**.
 - Evolved autonomous driving agents with **Genetic Algorithm + PPO reinforcement learning** to handle sparse rewards.
- Developed JavaScript music recommendation app with fine-tuned Huggingface LLM (**~500M params**).

Baker Lab, UT Austin – Undergraduate Research Assistant

(Apr 2024 – Aug 2024)

- Authored **\$100K** research grant, earned **4800 SU** compute grant for HPC distributed supercomputer.
- Developed [CLI plugin](#) with ChatGPT that **enhanced productivity**, reduced learning curve via natural-language interface.

Emerging Diagnostic and Investigative Technologies (EDIT) Lab – ML Intern

(Jun 2021 – Aug 2021)

- Developed Convolutional (**CNN**) tumor classifier, Graph Neural Network (**GNN**) to remove ink markings from H&E images.
- Performed **spatial domain analysis** on 16GB multimodal H&E + RNAseq data with Graph Convolution (**GCN**).

Publications

Adversarial License Plate Rim

(Aug 2025 – Jun 2026)

- Adversarial border degrades license plate reader accuracy by 60%. **First author publication:** <https://arxiv.org/abs/2604.02457>, **accepted** to Security in Machine Learning Applications workshop at ACNS conference 2026.
- Self-initiated:** sourced collaborators, managed cloud infrastructure across three providers, and wrote manuscript independently.

Blockchain Strategies for Real-World Governance

(May 2025 – Aug 2025)

- Can proof-of-stake voting improve governance? **First author preprint:** <https://ssrn.com/abstract=5393983>, **SSRN top paper**.